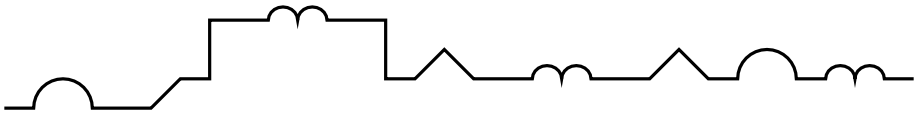
Math Kangaroo Practice

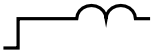
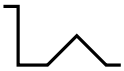
Grade 3-4 · 20 questions · harder · seed 32415

Part A: Each question is worth 3 points

1. This is the outline of a town seen from the river at sunset.



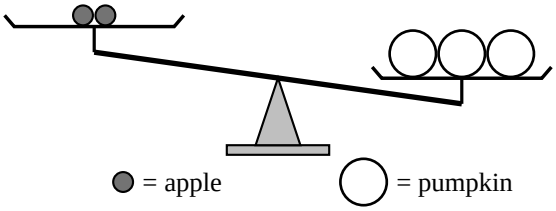
Which piece **cannot** be part of the outline?

A.  ***B.*** ***C.*** ***D.***  ***E.***

2. The odometer of a car shows 174938 km. All the digits of this number are different. After how many more kilometres will all the digits be different again?

A. 21 ***B.*** 13 ***C.*** 12 ***D.*** 11 ***E.*** 1

3. On one plate of a balance there are 2 apples and on the other there are 3 pumpkins. When we put 2 more pumpkins, exactly like the others, on the plate with the apples, the balance is level.



One pumpkin weighs the same as how many apples?

A. 5 apples ***B.*** 4 apples ***C.*** 3 apples ***D.*** 2 apples ***E.*** 1 apple

4. Emma, Iris, Felix and Anna stand in a row for a photo. Each time they stand in a different order. How many different photos can they take?

A. 4 ***B.*** 6 ***C.*** 9 ***D.*** 16 ***E.*** 24

Part B: Each question is worth 4 points

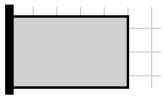
5. Aisha, Mia, Julien, Chloe and Ivan were born in different months: April, March, December, January and July. Mia was born in a month that starts with J. Ivan has a birthday in summer. Julien was born in winter. Aisha was not born in March.
Which statement is true?

A. Mia was born in July ***B.*** Aisha was born in March ***C.*** Mia was born in January
D. Chloe was born in January ***E.*** Ivan was born in January

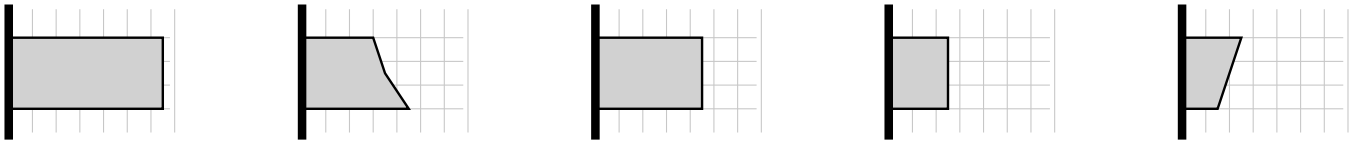
6. Finn has a rope 3 m long and cuts it into pieces that are each 20 cm long.
How many pieces does Finn get?

A. 15 ***B.*** 14 ***C.*** 13 ***D.*** 12 ***E.*** 10

7. From a window, Sara sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Sara draws the shadow of the flag on the wall at five different moments. Flat, the flag is 5 squares wide and 3 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat



A B C D E

How many of the five drawings can be right?

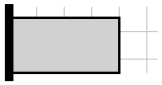
A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

8. Tom has the cards 1, 2, 4 and 6. Tom must use every card exactly once, with +, –, ×, ÷ and brackets.
Which number can Tom *not* get?

A. 44 ***B.*** 46 ***C.*** 50 ***D.*** 52 ***E.*** 53

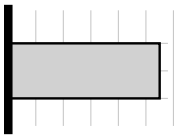
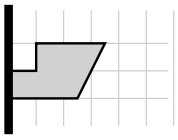
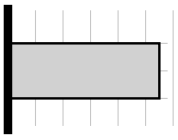
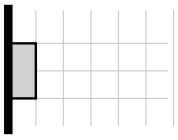
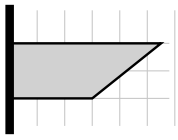
Part C: Each question is worth 5 points

9. From a window, Leo sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Leo draws the shadow of the flag on the wall at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Four of the drawings are wrong.



the flag, flat

Which drawing can be right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

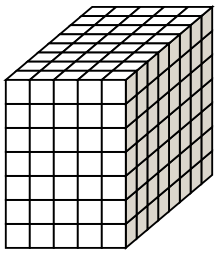
10. Nina has the cards 1, 3, 4 and 8. Using every card exactly once, with +, –, ×, ÷ and brackets, Nina wants to get the biggest possible number.
What is the biggest number Nina can get?

A. 128 ***B.*** 141 ***C.*** 155 ***D.*** 159 ***E.*** 160

11. Five children, Iris, David, Grace, Anna and Clara, sit in a row of 5 chairs at the cinema. Iris and David are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 120 ***B.*** 72 ***C.*** 50 ***D.*** 48 ***E.*** 24

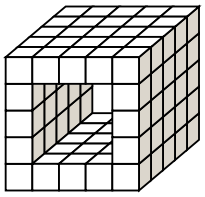
12. Chris builds a brick 5 cubes long, 8 cubes deep and 7 cubes tall. To make it strong, the red outside is **two** cubes thick everywhere (also at the bottom, top, front and back). Everything inside that is blue.



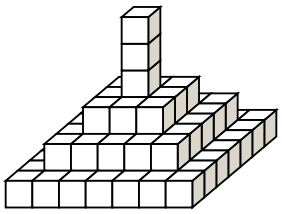
How many blue cubes are there?

A. 268 ***B.*** 90 ***C.*** 24 ***D.*** 12 ***E.*** 4

13. Nina has the tunnel on the left. Nina takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.



tunnel

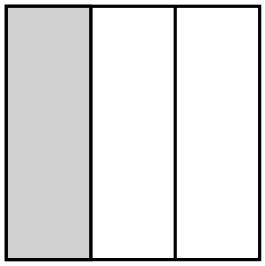


pyramid

How many more cubes does Nina need?

A. 70 ***B.*** 22 ***C.*** 7 ***D.*** 6 ***E.*** 4

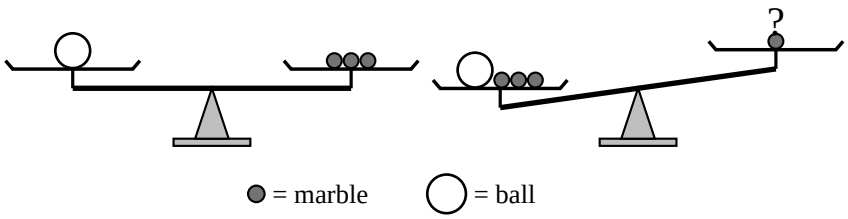
14. Mia cuts a square chocolate bar into 3 equal strips as shown. The perimeter of each strip is 72 cm.



How long is the side of the square?

A. 9 cm ***B.*** 18 cm ***C.*** 27 cm ***D.*** 36 cm ***E.*** 45 cm

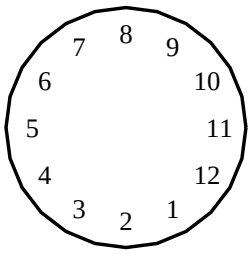
15. The first balance is level. On the second balance there is 1 ball and 3 marbles on the left and 1 marble on the right.



How many marbles must be added to the right plate to make it level?

A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 5

16. A round cake with the numbers 1 to 12 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 7 and 8, and between 2 and 3

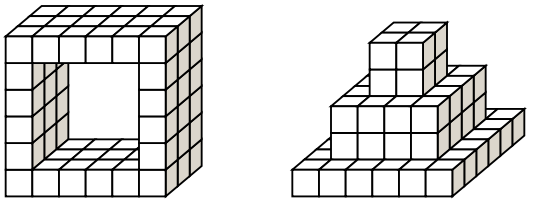
B. between 9 and 10, and between 3 and 4

C. between 11 and 12, and between 1 and 2

D. between 7 and 8, and between 5 and 6

E. between 11 and 12, and between 5 and 6

17. Robert has the tunnel on the left. Robert takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.

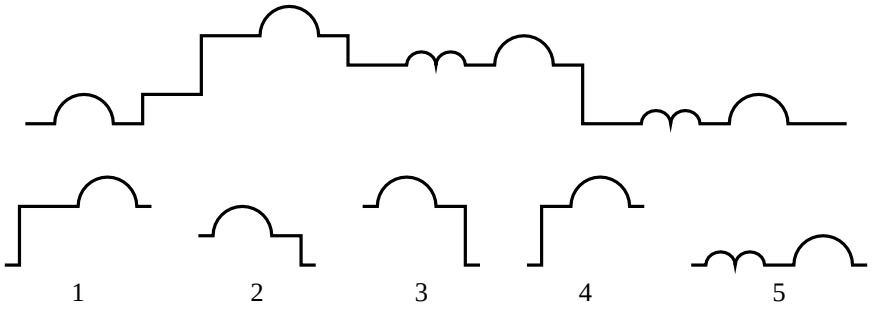


tunnel pyramid

How many more cubes does Robert need?

A. 56 ***B.*** 36 ***C.*** 19 ***D.*** 17 ***E.*** 16

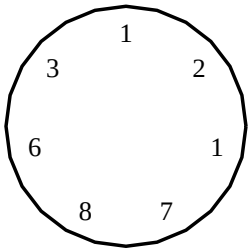
18. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 3 ***D.*** 4 ***E.*** 5

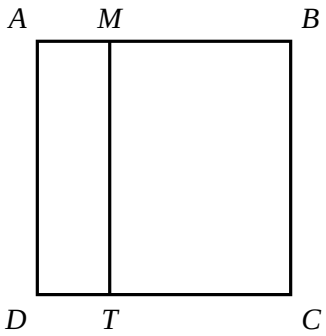
19. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

20. The square *ABCD* is cut into two rectangles. The perimeter of *AMTD* is 18 cm and the perimeter of *MBCT* is 24 cm.



What is the perimeter of the square *ABCD*?

A. 28 cm ***B.*** 21 cm ***C.*** 14 cm ***D.*** 7 cm ***E.*** 6 cm

Answer Key

1.	**A**	**2.**	**C**	**3.**	**D**	**4.**	**E**	**5.**	**C**	**6.**	**A**
7.	**E**	**8.**	**E**	**9.**	**D**	**10.**	**A**	**11.**	**D**	**12.**	**D**
13.	**D**	**14.**	**C**	**15.**	**E**	**16.**	**B**	**17.**	**E**	**18.**	**D**
19.	**B**	**20.**	**A**								

Solutions

1. A — All the other pieces can be found in the line, but this one cannot: its step is shorter than in the line. [2002 Q7 · cannot_belong]

2. C (12) — Counting up, every number before 174950 repeats a digit. 174950 has all different digits: 174950 – 174938 = 12. [2002 Q24 · all_different]

3. D (2 apples) — Take 2 pumpkins off both plates: 2 apples balance 3 – 2 = 1 pumpkin. So one pumpkin = 2 apples. [2002 Q9 · one_more_balances]

4. E (24) — 4 children can stand first, then 3 are left for the second place, and so on: 4 × 3 × 2 × 1 = 24. [2002 Q22 · photo_line]

5. C (Mia was born in January) — The only possible matching: Aisha – April; Mia – January; Julien – December; Chloe – March; Ivan – July. [2002 Q18 · which_true]

6. A (15) — 3 m = 300 cm. 300 ÷ 20 = 15. [2003 Q10 · pieces]

7. E (4) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 5 squares from the pole. Wrong: A. So 4 can be right. [2002 Q15 · count_possible]

8. E (53) — (2 × 6 – 1) × 4 = 44; (4 × 6 – 1) × 2 = 46; (1 + 4 × 6) × 2 = 50; (1 + 2 × 6) × 4 = 52. But with 1, 2, 4 and 6 nothing gives 53 (the closest are 52 = (1 + 2 × 6) × 4 and 54 = (1 + 2 × 4) × 6). [2002 Q21 · which_cannot]

9. D — Check two things: the pole edge must be covered all the way, and nothing may reach more than 4 squares from the pole. Only picture D passes both checks. [2002 Q15 · only_one_right]

10. A (128) — (1 + 3) × 4 × 8 = 128. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]

11. D (48) — Think of Iris and David as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block Iris and David can swap, so 2 × 24 = 48. [2002 Q22 · side_by_side]

12. D (12) — Two layers come off at each end of every length, so each length drops by 4: 1 × 4 × 3 = 12. [2003 Q9 · thick_wall]

13. D (6) — Tunnel: 16 × 5 = 80 cubes. Pyramid: 7 × 7 + 5 × 5 + 3 × 3 + 1 × 1 × 3 = 86. Missing: 86 – 80 = 6. [2002 Q20 · not_enough]

14. C (27 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 72 ÷ 8 = 9, so the side is 3 × 9 = 27 cm. [2002 Q6 · strips_to_square]

15. E (5) — One ball = 3 marbles, so the left plate is worth 1 × 3 + 3 = 6 marbles. The right plate has 1, so add 6 – 1 = 5. [2002 Q9 · what_to_add]

16. B (between 9 and 10, and between 3 and 4) — Total 78, so each piece has 39: 10+11+12+1+2+3 = 39 and 4+5+6+7+8+9 = 39. [2002 Q11 · straight_crack]

17. E (16) — Tunnel: 20 × 3 = 60 cubes. Pyramid: 6 × 6 + 4 × 4 × 2 + 2 × 2 × 2 = 76. Missing: 76 – 60 = 16. [2002 Q20 · not_enough]

18. D (4) — Pieces 1, 2, 3, 5 fit. Not: 4 (it is a mirror image of a part of the line). [2002 Q7 · how_many_fit]

19. B (1) — Each half must have 14. The ways: 8+6 | 3+1+2+1+7. [2002 Q11 · count_ways]

20. A (28 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 18 + 24 = 42 cm = 6 sides. Side = 42 ÷ 6 = 7 cm, perimeter 4 × 7 = 28 cm. [2002 Q6 · two_perimeters]